MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, LOW NOISE OPERATIONAL
AMPLIFIERS, MONOLITHIC SILICON

This amendment forms a part of Military Specification MIL-M-38510/131, dated 15 July 1982, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.1 line 2: Delete "Three" and substitute "Two".

PAGE 3

TABLE I, dV_{IO}/dt, limits column: Delete "±8" and substitute "±10". Make this substitution in table III, test nos. 24 and 44.

TABLE I, CMRR, conditions column: Delete "V_{CM} = ±12V" and substitute "V_{CM} = ±11V".

TABLE I, $I_{OS}(+)$, min. limits column: Delete "-70" and substitute "-95". Make this substitution in table III, test nos. 17 and 37.

TABLE I, $I_{OS}(-)$, max. limits column: Delete "70" and substitute "95". Make this substitution in table III, test nos. 18 and 38.

PAGE 4

TABLE I, delete and substitute:

Slew rate	SR(+) and SR(-)	V_{IN} = 10 volts	01	10	--	V/µs
			02	3	--	V/µs

Make this change in table III, test nos. 100, 101, 107, 108, 111, and 112.

TABLE I, EN, max. limits column: Delete "11" and substitute "15". Make this substitution in table III, test no. 70.

PAGE 5

TABLE I, delete and substitute the following:

"

Input noise current density	IN	f_o = 30 Hz (see fig. 10)	--	10	$pA/\sqrt{Hz}$
		f_o = 100 Hz	--	5	$pA/\sqrt{Hz}$
		f_o = 1 kHz	--	2	$pA/\sqrt{Hz}$
		f_o = 10 kHz	--	2	$pA/\sqrt{Hz}$

"

Make these changes in table III test nos. 96, 97, 98, and 99.

FSC 5962

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TABLE I, C_S, min. limits column: Delete "100" and substitute "80". Make this change in table III, test nos. 103 and 104.

NOTE 2/, line 2: Delete "7K" and substitute "2K".

PAGE 6

Para 3.5: Delete second sentence in its entirety.

TABLE II: Delete "Class C devices" column in its entirety.

PAGE 15

FIGURE 8: Add note, "6/" C_L includes scope, probe, and jig capacitance". Make this addition as note 7 to figure 9 on page 16.

PAGE 17

FIGURE 10: Delete and substitute new figure 10 on page 4 of this amendment.

PAGE 18

FIGURE 11: Delete and substitute new figure 11 on page 5 of this amendment.

PAGE 19

FIGURE 12: Add to the right side of each 22pF capacitor "3/" and add note to read as follows, "3/ Device type 01 only".

PAGE 20

TABLE III, test no. 1, adapter pin nos. 1, 2, and 3: Delete "27, -3, -12" and substitute "26, -4, -11" respectively. Make these changes to test nos. 5, 8, 20, 25, 28, 40, 45, and 48.

TABLE III, test no. 2, adapter pin nos. 1, 2, and 3: Delete "3, -27, 12" and substitute "4, -26, 11" respectively. Make these changes to test nos. 7, 10, 21, 27, 30, 41, 47, and 50.

TABLE III, test nos. 15 and 16, notes column: Add "01 only".

TABLE III, test no. 18, adapter pin number 3: Delete "-10" and substitute "10".

PAGE 23

TABLE III, test nos 70 and 71, both limits columns: Delete "nv/Hz" and substitute "$nv/\sqrt{Hz}$". Make these same changes to test nos 94 and 95.

PAGE 24

TABLE III, test no. 96, equations column: Delete "$IN=(((91)^2 - (61)^2 - 1644)/10000)\cdot 5$" and substitute "$IN=(((E91)^2 - (E61)^2 - 1644)/10000\cdot 5$".

TABLE III, test no. 97, equations column: Delete "$IN=(((92)^2 - (89)^2 - 1644)/10000)\cdot 5$" and substitute "$IN=(((E92)^2 - (E89)^2 - 1644)/10000\cdot 5$".

TABLE III, test no. 98, equations column: Delete "$IN=(((93)^2 - (62)^2 - 1644)/10000)\cdot 5$" and substitute "$IN=(((E93)^2 - (E62)^2 - 1644)/10000\cdot 5$".

TABLE III, test no. 99, equations column: Delete "$IN=(((94)^2 - (90)^2 - 1644)/10000)\cdot 5$" and substitute "$IN=(((E94)^2 - (E90)^2 - 1644)/10000\cdot 5$".

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PAGE 26

4.2.d: Delete and substitute the following, "The percent defective allowable (PDA) shall be as specified in MIL-M-38510".

PAGE 27

4.4.3.1, line 3: Delete "and C".

Custodians:
Air Force - 17
Army - ER
Navy - EC
NASA - NA

Review activities:
Air Force - 11, 19, 80, 85, 99
Army - AR, MI
Navy - OS, SH
DLA - ES

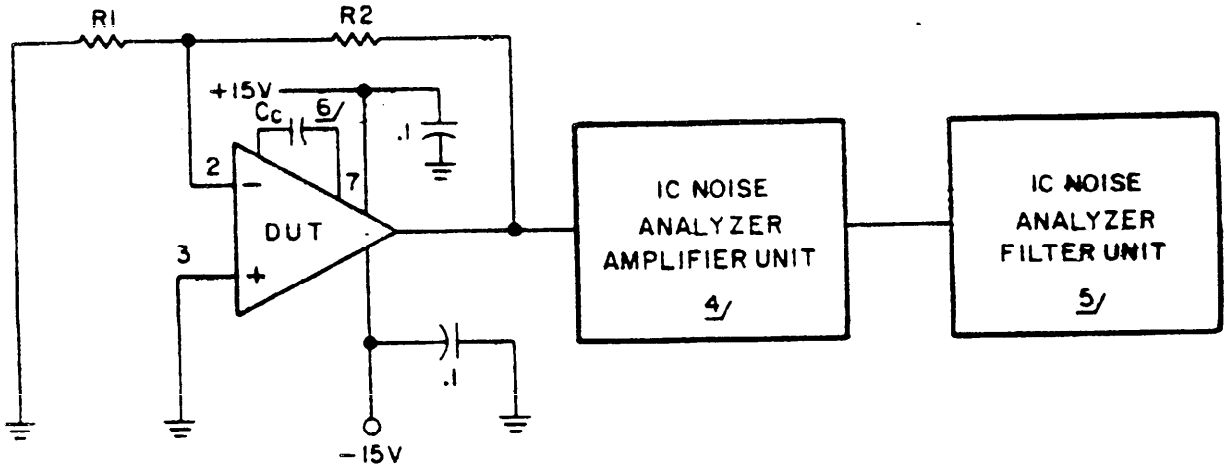
User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
Air Force - 17

(Project 5962-0688)

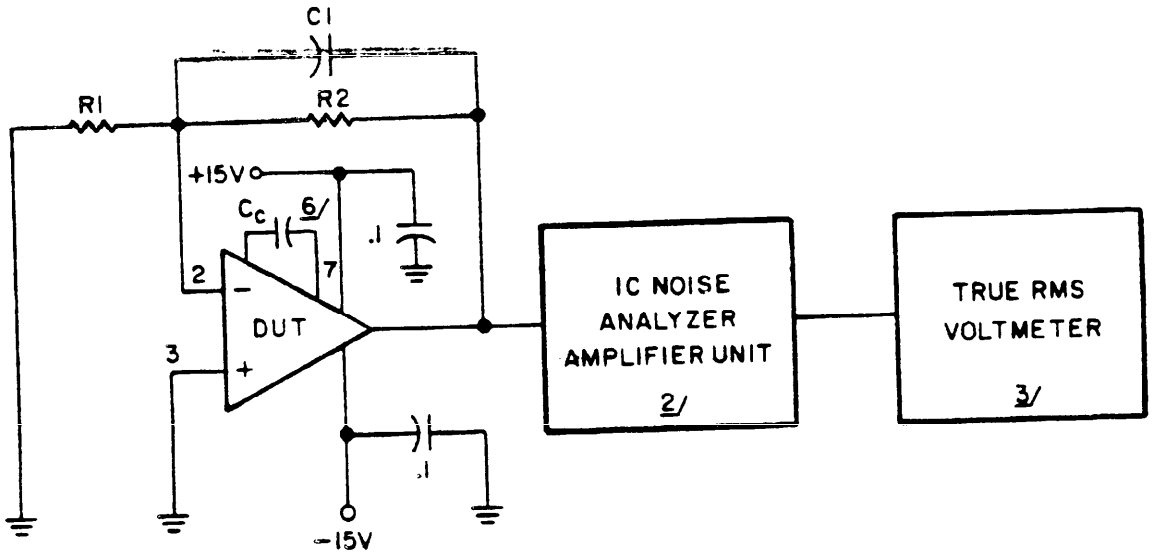
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NOTES:
1. Input Noise Voltage Density (En) Test: R1 = 50 Ω, R2 = 10 kΩ.
2. Input Noise Current Density (In) Test: R1 = 105 Ωk, R2 = 2.0 MΩ.
3. All resistors are metal film and ±1% tolerance. Capacitors are in microfarads and are ± 10% tolerance.
4. Quan-Tech Model 2283 or equivalent.
5. Quan-Tech Model 2181 or equivalent.
6. C_c = 20 pF, device type 01 only.

FIGURE 10. Noise density test circuit.

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NOTES:
1. All resistors are metal film and are ±1% tolerance. Capacitors are in microfarads and are ±10% tolerance.
2. Quan-Tech Model 2283 equivalent. Bandwidth must be 10 Hz to 20 kHz minimum. Cain = 50.
3. Bandwidth must be 10 H_z to 20 kH_z minimum.
4. Effective circuit noise banswidth is 10 kH_z.
5. R1 = 50 Ω, R2 = 10 kΩ, C1 = 2500 pF.
6. C_C = 22 pF, device type 01.

FIGURE 11. Broadband noise test circuit.

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